INTRODUCTION TO INFORMATION GEOMETRY

Payoff Matrices and Expected Fitness

Week 2 · Lecture 1

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Fitness comes from encounters

In a well-mixed population, a type meets A with probability x and B with probability $1 - x$.

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

	Opponent A	Opponent B
Type A receives	a	b
Type B receives	c	d

A weighted average of encounters

An A individual receives payoff a from an A opponent and payoff b from a B opponent.

$$f_A(x) = ax + b(1 - x).$$

Similarly,

$$f_B(x) = cx + d(1 - x).$$

Both expected fitnesses depend on the current population composition.

A matrix becomes a dynamical model

Let $M = \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$.

$$f_A(x) = 2x + 4(1-x) = ?, \qquad f_B(x) = 3x + (1-x) = ?.$$

Find the fitnesses at $x = \frac{1}{4}$.

$$f_A(\tfrac{1}{4}) = ?, \qquad f_B(\tfrac{1}{4}) = ?.$$

A matrix becomes a dynamical model

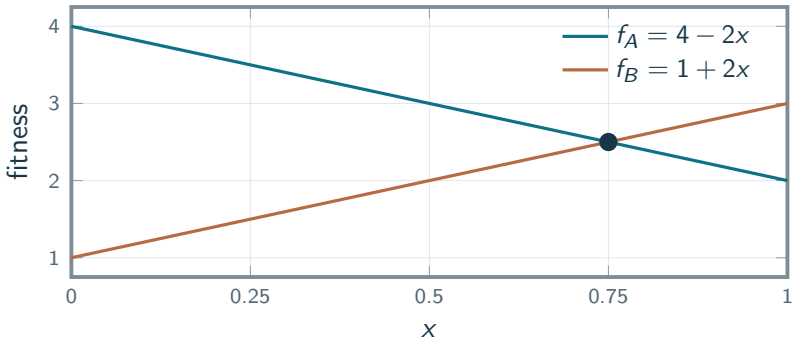
Let $M = \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$.

$$f_A(x) = 2x + 4(1-x) = 4 - 2x, \qquad f_B(x) = 3x + (1-x) = 1 + 2x.$$

At $x = \frac{1}{4}$,

$$f_A(\tfrac{1}{4}) = \tfrac{7}{2}, \qquad f_B(\tfrac{1}{4}) = \tfrac{3}{2}.$$

The two expected fitness functions



Where the two lines intersect, both types have equal fitness.

The advantage is an affine function

Subtract the fitnesses:

$$\Delta(x) = f_A(x) - f_B(x)$$

$$= \underbrace{(b-d)}_{\Delta(0)} + \underbrace{(a-b-c+d)}_{\text{slope}}\, x.$$

The differential equation is

$$x' = x(1-x)\Delta(x).$$

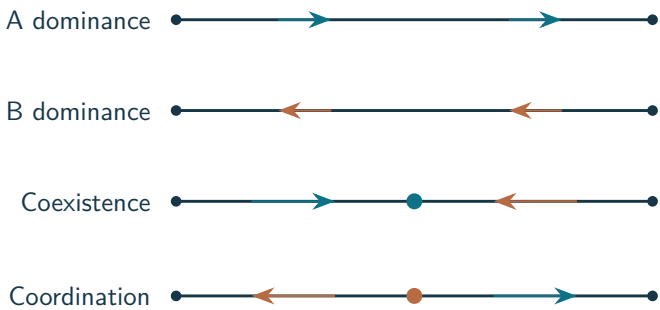
Reading the advantage at the endpoints

$$\Delta(0) = b - d, \qquad \Delta(1) = a - c.$$

- At $x \approx 0$, a rare A mostly meets B. A increases when $b > d$.
- At $x \approx 1$, a rare B mostly meets A. B increases when $c > a$.

These invasion tests determine the direction of motion near the two ends.

Four phase portraits from two signs



The signs of $\Delta(0)$ and $\Delta(1)$ determine the case.

A coordination example

Consider $M = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$. Find $\Delta(0)$, $\Delta(1)$, and the interior equilibrium.

$$\Delta(0) = ?, \qquad \Delta(1) = ?, \qquad x^* = ?.$$

Which endpoint does a solution approach from $x(0) = 0.35$?

A coordination example

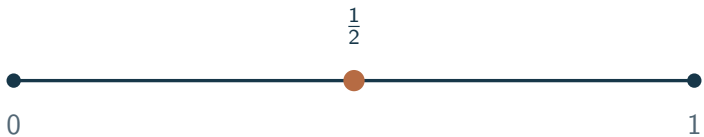
Consider $M = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$. The advantage is $\Delta(x) = 4x - 2$.

$$\Delta(0) = -2, \quad \Delta(1) = 2, \quad x^* = \tfrac{1}{2}.$$

The solution from $x(0) = 0.35$ approaches 0.

The coordination phase line

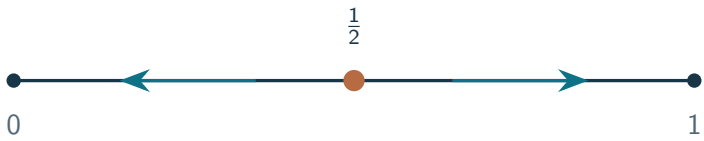
For $x' = x(1-x)(4x-2)$, the interior equilibrium is $x^* = \frac{1}{2}$.



Draw the arrows. Determine the basins of attraction.

The coordination phase line

For $x' = x(1-x)(4x-2)$, the interior equilibrium is $x^* = \frac{1}{2}$.



The basins are $[0, \frac{1}{2})$ and $(\frac{1}{2}, 1]$.

The one-dimensional conclusion

For a two-strategy payoff matrix,

$$x' = x(1-x)\big[(b-d) + (a-b-c+d)x\big].$$

The difference between the rows determines the motion.

- Its values at 0 and 1 determine the phase portrait.
- Its zero, when interior, locates the coexistence or threshold state.